Zhiming Xu

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https://github.com/Mr-Wonderfool

EDUCATION

Tongji University 09/2022-07/2026 (Expected)

- Bachelor of Engineering in Computer Science, GPA：4.85/5.00 (Rank Top 5% in honored class, Guohao Academy).

RESEARCH EXPERIENCE

Robotics and Artificial Intelligence Lab, Tongji University 03/2024-Present

1. **Zhiming Xu** et al, *KinematicRL: Bridging Sim-to-Real Gap With Model-Based Demonstrations and Domain Randomization*. IEEE Transactions on Automation Science and Engineering (T-ASE 2025, Under Review). Deep reinforcement learning social navigation framework integrating dynamic constraints, achieving human tracking via 2D LiDAR and crowd adaptation through attention mechanisms, enabling zero-shot sim-to-real transfer.
2. **Zhiming Xu** et al, *Principled Latent Space For Dynamics-Invariant Fast Adaptation*. (Ongoing Research) Propose a variational framework that learns latent representations of robot dynamics from trajectory histories, jointly optimized with actor-critic objectives to learn a family of policies that are capable of adapting to different dynamics.

Research Interests: Deep Reinforcement Learning, Robotics, Motion Planning, Vision Language Models.

INDUSTRY EXPERIENCE

Multimodal Agent Group, Autonomous Driving Divison, NIO Inc. 07/2025-11/2025

Project I: Learning From Multimodal Demonstrations With Auto-Regressive Architecture

1. **Vocabulary Construction**: Develop a custom clustering algorithm to extract 384 representative motion tokens from 20k driving clips, under which trajectory reconstruction error stabilizes around 0.06m regardless of trajectory length. Utilize top-k selection to inject controlled noise during tokenization to increase robustness of the model.
2. **Model Design**: Construct a query-based architecture, where ego query and generated tokens attend, via deformable attention, to BEV features, then interact with context queries to autoregressively generate variable-length path.

Project II: Framework For Fine-tuning Auto-Regressive Models With Reinforcement Learning

1. **Planner RL Integration**: Integrate RL into auto-regressive planners via a custom parallel fine-tuning pipeline (implement PPO/GRPO/GSPO on model built with *mmdet3d*), reducing collision rate of baseline model by 18%.
2. **Address Real-World Issues**: For problems such as proximity to obstacles and kinematically infeasible trajectories, use tailored reward functions (e.g., proximity and curvature penalty) to bring key metrics into acceptable ranges.

Project III: Research of Vision-Language Navigation Foundational Model

1. **Dual-System Design**: Building upon *InternVLA-N1*, establish an asynchronous structure consisting of a VLM (slow system) and an action expert (fast execution), further incorporating a plan token to pass VLM-generated pixel-goal information to action expert, enabling joint training of two systems.
2. **Evaluation on Wheeled Robot**: Construct in-door navigation scenarios based on real autonomous driving contexts to evaluate scene understanding and instruction following, laying the foundation for future VLA system development.

SELECTED PROJECTS

Motion Planning with Deep Reinforcement Learning 03/2025-Present

1. Constructed a lightweight 2D LiDAR-based navigation simulator, achieving ~50× faster training efficiency than Gazebo simulator through vectorized sensor simulation and data collection.
2. Implemented classic DRL algorithms (SAC, PPO, TD3, DQN, etc) in a plug-and-play training framework.
3. Supported dynamic/static obstacles with arbitrary shapes, customizable robot geometries, and integrated domain and dynamics randomization to enhance policy robustness.
4. Successfully zero-shot transferred the trained policy to Gazebo simulator and, combined with a global planner, built a hierarchical system integrating collision avoidance and long-range goal-selection in complex indoor environments.

Reinforcement Learning for Enhancing Chain-of-Thought Consistency 07/2025-08/2025

1. Constructed high-quality chain-of-thought supervision data on the GQA dataset and proposed a consistency metric that detects reward hacking in Visual Question Answering (VQA) by verifying the alignment between reasoning keywords and final answers.
2. Employed a two-stage training framework: (1) Implemented SFT to fine-tune *Qwen2.5-VL-3B-Instruct* for structured reasoning outputs; (2) In the RL stage (built upon *trl+vLLM*), designed reward functions (answer correctness, format compliance, reasoning–answer consistency) and compared GRPO vs. GSPO for reasoning quality improvement.
3. Achieved +28.6% accuracy and +11.7% chain-of-thought consistency gains on GQA using GSPO. [Repository]

SELECTED AWARDS

- **National Scholarship** x2 (Highest scholarship awarded by Chinese Government, top 0.1%) 09/2023, 09/2025
- **Alishan Scholarship** (Sole undergraduate recipient in Tongji University) 09/2024
- **National First Prize** in National Mathematical Contest in Modeling (Awarded to top 0.5% teams) 11/2024
- **National First Prize** in National English Debating Competition (National Best Speaker Award) 06/2024

SERVICES

Captain & Mentor, Tongji English Debating Team 03/2024-Present

- Led the university's top debating team and devoted extensive hours to recruiting, training, and mentoring a new generation of 20+ debaters in logical reasoning and public speaking.

Peer Mentor, Guohao Academy 06/2024-Present

- Voluntarily mentored over 100 junior students on academic planning, major selection, and personal development, fostering a supportive peer community.

SKILLS & OTHERS

- Programming Languages: Python, C/C++, MATLAB, JavaScript
- Robotics Related: ROS, MuJoCo
- Others: PyTorch, huggingface transformers, mmdet/mmcv, vim, git, cmake
- Language Proficiency: TOEFL(110), GRE(325+5)